Faculty of Science, Engineering and Built Environment

SET111 Sustainable Design

Deakin University Unit Guide

Trimester 1, 2025

CONTENTS

Welcome 2
Who is the unit team? 2
Unit chair: leads the teaching team and is responsible for overall delivery of this unit 2
Other members of the team and how to contact them 2
Administrative queries 2
About this unit 2
Unit development in response to student feedback 2
Learning Outcomes 3
Assessing your achievement of the unit learning outcomes 3
Hurdle requirements 3
Summative assessment task 1 4
Summative assessment task 2 5
Your learning experiences in this unit 6
Educator-facilitated (scheduled) learning activities - on-campus unit enrolment 6
Educator-facilitated (scheduled) learning activities - online unit enrolment 6
Typical study commitment 6
Note 7
Unit learning resources 7
Essential learning resources 7
Recommended learning resources 7
Where to access unit resources 8
Key dates for this study period 8
Unit weekly activities 8

Welcome

This unit focuses on the principles of design and the use of computer-aided design (CAD) to develop solutions that reduce the environmental impact of artefacts in their production and/or use. Students will engage with sustainable design strategies and practices, learning how to create innovative design solutions that address environmental considerations. The unit also emphasises the importance of adhering to international and Australian standards in engineering design, fostering the development of professional documentation skills. By critically evaluating and refining design solutions, students will acquire the discipline specific technical expertise and professional practice skills, preparing them for a career in engineering with a strong emphasis on sustainability.
This Unit Guide provides you with the key information about this unit. Please read it carefully and refer to it frequently throughout the study period. Your unit site also provides information about your rights and responsibilities. We will assume you have read this before the unit commences, and we expect you to refer to it throughout the study period.

To be successful in this unit, you must:

- read all materials in preparation for your learning activities and follow up each with further study and research on the topic
- start your assessment tasks well ahead of the due date
- read or listen to all feedback carefully and use it in your future work
- attend and engage in all educator facilitated (scheduled) learning activities and other learning experiences as part of the unit design

Who is the unit team?

Unit chair: leads the teaching team and is responsible for overall delivery of this unit

Name: Saleh Gharaie
Campus: Melbourne Burwood
Email: s.gharaie@deakin.edu.au
Phone: +61 3 522 73765
Other members of the team and how to contact them

Details of any additional team members will be provided on the unit site.
Administrative queries

- check-out the 'SEBE Student Hub' section on your unit site
- contact your Unit Chair or Campus Leader
- drop in or contact Student Central to speak with a Student Adviser

For additional support information, please see the Rights and Responsibilities section under 'Content' in your unit site.
About this unit

Unit development in response to student feedback

Every trimester, we ask students to tell us, through eVALUate, what helped and hindered their learning in each Unit. You are strongly encouraged to provide constructive feedback for this Unit when eVALUate opens (you will be emailed a link).
In previous versions of this unit, students have told us that these aspects of the Unit have helped them to achieve the learning outcomes:

- Practical CAD worksheets that allow students to apply and refine their skills incrementally.
- Step-by-step CAD video tutorials, which provide accessible guidance for learning Fusion.
- Weekly studio sessions that facilitate one-on-one interactions and hands-on support from teaching staff.
- Clear rubrics and guides that outline assignment expectations and assist in self-directed learning.
- Opportunities for self-paced learning through online resources and recorded lectures.

- A strong focus on practical applications of CAD and engineering principles in portfolio tasks.

The following aspects of the Unit have been introduced, enhanced or retained in response to feedback from students who have undertaken this unit in previous trimesters:

- Enhanced content related to foundational concepts of sustainability, with a focus on providing more examples of Design for Sustainability to improve understanding and application.
- Additional examples for portfolio tasks to clarify expectations and help students manage workload effectively.
- Introduction of additional CAD worksheets aimed at better preparing students for the CAD skills test and reinforcing CAD skills development.
- Updated and modernised CAD tutorial videos to ensure alignment with course content and student needs.
- Improved in-class activities that focus on building core CAD skills while offering flexibility for students at different experience levels.
- Implementation of structured practice CAD tests to reduce stress and build confidence ahead of the CAD test.

If you have any concerns about the Unit during the trimester, please contact the unit teaching team - preferably early in the trimester - so we can discuss your concerns, and make adjustments, if appropriate.

Learning Outcomes

Each Unit in your course is a building block towards Deakin's Graduate Learning Outcomes - not all Units develop and assess every Graduate Learning Outcome (GLO).

ULO	These are the Unit Learning Outcomes (ULOs) for this unit. At the completion of this unit, successful students can:	Alignment to Deakin Graduate Learning Outcomes (GLOs)
ULO1	Apply principles of design to develop solutions that reduce the environmental impact of artefacts in their production and/or use.	GLO1: Discipline-specific knowledge and capabilities. GLO5: Problem solving.
ULO2	Apply computer-aided design (CAD) skills for design problems and technical documentation.	GLO1: Discipline-specific knowledge and capabilities GLO3: Digital literacy GLO5: Problem solving
ULO3	Synthesise and communicate complex design concepts and technical information through diverse visualisation and documentation techniques to convey technical information to a range of stakeholders.	GLO2: Communication GLO6: Self-management
ULO4	Utilise international and Australian standards to communicate engineering designs.	GLO1: Discipline-specific knowledge and capabilities GLO8: Global citizenship
ULO5	Conduct a critical evaluation of a sustainable design solution for an existing artefact, analysing its effectiveness in meeting sustainability objectives and project requirements.	GLO4: Critical thinking

Assessing your achievement of the unit learning outcomes

Hurdle requirements

To be eligible to obtain a pass in this unit, students must achieve an overall mark of at least 50% in CAD skills test in addition to the normal requirement of at least 50% overall for the unit.

Brief summary of the hurdle requirement/s	Rationale
To be eligible to obtain a pass in this unit, students must achieve an overall mark of at least 50% in CAD skills test in addition to the normal requirement of at least 50% overall for the unit.	The CAD skills test assesses students' ability to utilise essential 3D modelling tools. ULOs 2, 3 and 4 are linked to this assessment task. These ULOs represent skills that students are required to master early on in the course. To reflect this, students must demonstrate they are able to meet the minimum engineering standards for these assessment tasks described by the assessment guide.

Summative assessment (tasks that will be graded or marked)
NOTE: It is *your responsibility* to keep a backup copy of every assignment and the materials used to develop/complete it where possible (e.g. written/digital reports, essays, videos, images). In the unusual event that one of your submissions becomes corrupted, is incorrectly submitted or otherwise lost, you may be asked to submit the backup copy. Any work you submit may be checked by electronic or other means for the purposes of detecting breaches of academic integrity such as collusion, plagiarism and contract cheating. You must understand your responsibility to act with honesty and integrity in your studies as Deakin takes all breaches very seriously. Make sure you read Your rights and responsibilities as a student in this unit to find out more about academic integrity.

Deakin has a universal assessment submission time of 8 pm AEDT/AEST. A late penalty will apply to assessments submitted after 11.59 pm AEDT/AEST.
Summative assessment task 1

Title	Design Portfolio (Individual)
Brief description of assessment task	This individual assessment task is a design challenge requiring students to create a design portfolio by applying the principles of sustainable design and engineering practices.
Brief description of assessment task (continued)	Students are expected to apply foundational and advanced design principles, including computer-aided design (CAD) skills, to develop solutions that reduce the environmental impact of artefacts in their production and/or use. The developmental process involves analysing an existing artefact to identify opportunities for sustainability improvements, defining the problem, generating ideas, evaluating design alternatives, creating virtual models, and effectively communicating the final design through technical documentation and visualisation techniques. Students will also justify and critically evaluate their design solutions to ensure alignment with sustainability objectives and project requirements.
Detail of student output	The Design Portfolio assessment task comprises 10 tasks as follows: Part 1: 1. Problem definition 2. Mind map 3. Concept Sketches 4. Evaluation of Design alternatives Part 2: 5. Information Sketch 6. 3D generated Models 7. 2D technical drawings 8. 3D Rendered images 9. Infographic 10. Critical design review
Grading and weighting (% total mark for unit)	70% - marked and graded (Part 1 30%, Part 2 40%)

This task assesses your achievement of these Unit Learning Outcome(s)	**ULO1:** Apply principles of design to develop solutions that reduce the environmental impact of artefacts in their production or use. Assessed across Tasks 1–4, where students define the design challenge, propose sustainability-focused objectives, map out design strategies, generate concept sketches, and evaluate design alternatives. These tasks collectively demonstrate the application of design principles to address environmental impacts. **ULO2:** Apply computer-aided design (CAD) skills for design problems and technical documentation. Assessed in Tasks 6 and 7, where students create 3D CAD models and detailed 2D engineering drawings of their design. Proficiency in CAD software and adherence to technical documentation standards are evaluated. **ULO3:** Apply information, graphic, and spatial skills and techniques to convey technical information to a range of stakeholders. Assessed through Tasks 5, 6, 8, and 9. Students demonstrate their ability to effectively communicate technical information through detailed technical sketches (Task 5), 3D CAD models (Task 6), and 3D rendered images (Task 8), showcasing their design visually and clearly. Additionally, they create an infographic (Task 9) to summarise their project for a broader audience, focusing on clarity, organisation, and accessibility. These tasks collectively assess the student's ability to convey complex design concepts in a variety of format. **ULO4:** Utilise international and Australian standards to communicate engineering designs. Assessed through Task 7, where students produce detailed 2D technical drawings of their design. The task evaluates their adherence to international and Australian standards, such as **ULO5:** Conduct a critical evaluation of a sustainable design solution for an existing artefact, analysing its effectiveness in meeting sustainability objectives and project requirements. Assessed in Task 10, where students conduct a critical design review, reflecting on strengths, weaknesses, and alignment with sustainability objectives, while proposing feasible improvements.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1: Discipline-specific knowledge and capabilities essential for professional practice in engineering, including sustainable design and CAD skills. GLO2: Communication skills developed through the clear and effective presentation of technical information and design solutions to diverse stakeholders. GLO3: Digital literacy demonstrated through the use of CAD software and other digital tools to create and document design solutions. GLO4: Critical thinking applied in the analysis and evaluation of design alternatives and sustainable solutions. GLO5: Problem solving demonstrated through the development of innovative solutions to reduce the environmental impact of artefacts. GLO6: Self-management developed through the independent organisation and presentation of complex design information. GLO8: Global citizenship demonstrated through the application of international and Australian standards in design practices.
How and when you will receive feedback on your work	Students will receive feedback on their submissions for Part 1 within two teaching weeks of the submission date. Feedback for Part 2 will be provided alongside the final overall grade, which will be released on Thursday, 3 July. Feedback will be given based on the assessment rubric.
When and how to submit your work	The portfolio will be broken into two submissions: 1. Design portfolio assessment task-part 1 (Tasks 1-4) due by 8pm (AEDT) on Friday, 4 April 2025 in Week 5 2. Design portfolio assessment task-part 2 (Tasks 5-10) due by 8pm (AEST) Monday, 26 May in Week 12 (Study period) Full task requirements can be found on the unit site.

Summative assessment task 2

Title	CAD Skills Test (Individual)
Brief description of assessment task	This assessment task is based on your ability to create a 3D part model and a single engineering drawing under test conditions. Autodesk Fusion is the only software that can be used.
Detail of student output	The CAD test requires the creation of a 3D model with accurate dimensions and constraints, incorporating features specified in the test guidelines You are also expected to produce engineering drawings that adhere to the specifications outlined in the test. These drawings must include all critical dimensions as listed in the marking rubric.
Grading and weighting (% total mark for unit)	30% - marked and graded
This task assesses your achievement of these Unit Learning Outcome(s)	**ULO2:** Apply computer-aided design (CAD) skills for design problems and technical documentation. Assessed through the creation of a 3D part model and a 2D engineering drawing during the CAD Skills Test. Students demonstrate their proficiency in CAD software by producing accurate, constrained models and detailed technical drawings that effectively document their design. **ULO3:** Apply information, graphic, and spatial skills and techniques to convey technical information to a range of stakeholders. Assessed through the 2D engineering drawing, where students communicate technical details clearly and professionally. The drawing must include all critical dimensions and annotations, ensuring it can be interpreted by various stakeholders. **ULO4:** Utilise international and Australian standards to communicate engineering designs. Assessed through the 2D engineering drawing, which must adhere to Australian and international standards, such as AS1100. Compliance with these standards is evaluated to ensure students can produce professional and standardised documentation.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1: Discipline-specific knowledge and capabilities: By applying CAD skills to create a 3D model of an artefact. GLO2: Communication: By producing 2D technical drawings to communicate complex design details effectively.
How and when you will receive feedback on your work	Students will receive feedback within two weeks of the submission date. Feedback will be given based on the assessment rubric.
When and how to submit your work	To be held in student's allocated studio sessions in Week 10. Further details will be provided on the unit site.

Your learning experiences in this unit

Educator-facilitated (scheduled) learning activities - on-campus unit enrolment

1 x 2 hour practical experience (studio) per week.

Educator-facilitated (scheduled) learning activities - online unit enrolment

Online independent and collaborative learning including a 1 x 2 hour online practical experience (studio) per week.

Typical study commitment

Students will on average spend 150 hours over the teaching period undertaking the teaching, learning and assessment activities for this unit.

This will include educator guided online learning activities within the unit site.

Learning and assessment activities in this unit for both campus and online students includes watching pre-recorded content, designated activities in the studio, assessment tasks, readings and study time.

Learning activities will engage students in locating and collecting information, critical thinking and analysis, and application of design strategies and modelling techniques to develop practical solutions to real-world environmental problems. Students will gain experience through practice; therefore, self-directed study time should be allocated for discovery and integration of the design principles with design process.

Note

At Deakin, courses are delivered within a learning environment that provides all students with equitable and consistent access to facilities, infrastructure, resources, and support to assist student progress and achievement of learning outcomes.

We have introduced new terms to reflect learning activities to enhance your learning experience, aligning with our innovative **DeakinDesign learning principles and practices**. The new terms better reflect how teaching teams will guide you through your learning journey and the types of learning experiences you will have.

'Lectures' are the activities where teaching staff engage you through presentations with student participation.

In **'seminars'**, an educator will guide you in a smaller group of students through highly interactive discussions and activities.

Your units may also include '**practical experiences**' such as '**laboratory**', '**workshops**', '**clinical skills**' and more. These hands-on activities typically take place in specialised facilities with industry tools, equipment or technology to allow you to apply your knowledge practically.

Some other terms

If you see a '**meeting**' in your timetable, this is an optional drop-in session.

'Assessments' or **'team-based learning**' indicate an in-class evaluation of your skills or knowledge. A **'pre-assessment practice'** could be scheduled to prepare you for these assessments.

Find out more

Take a look at the **Learning activities webpage** for a full list of the terminology changes and reasons they were changed.

Unit learning resources

Your unit learning resources can be accessed from your unit site.
The texts and reading list for SET111 can be found via the University Library.

Note: Select the relevant trimester reading list. Please note that a future teaching period's reading list may not be available until a month prior to the start of that teaching period so you may wish to use the relevant trimester's prior year reading list as a guide only.

Essential learning resources

Learning resources for this unit will be available to students via the Unit site. These resources include:

- Audio-visual and reading materials.

Software required for this unit will be provided free of charge and will be accessible from Week 1.

Recommended learning resources

The Deakin Software Library provides students with access to software that you may need or find useful for your study at Deakin.

The following textbooks are recommended for further reading and are available in the University Library:

- Sustainable Product Design and Development: Desal, A 2020. Taylor & Francis Group Milton
- Engineering drawing: Boundy, AW 2001. McGraw-Hill Sydney
- Engineering drawing: the leading Australian textbook for students of engineering drawing and graphics: Boundy, AW 2015. McGraw-Hill Sydney.

Where to access unit resources

Textbooks can be sourced from various outlets including direct from the publisher, online bookshops, or retailers. Limited copies of textbooks are also available on loan from the University Library.

Key dates for this study period

Trimester 1 teaching period begins	Monday 3 March 2025
Census date	Monday 31 March 2025
Easter/intra-trimester break	Friday 18 April - Sunday 27 April 2025
Trimester 1 teaching period ends	Friday 23 May 2025
Study period (end-of-unit assessment /examination preparation period)	Monday 26 May - Friday 30 May 2025
End-of-unit assessment and examination period	Monday 2 June - Friday 13 June 2025
Inter-trimester break (the period between trimesters)	Monday 16 June - Friday 4 July 2025
Unit results released	Thursday 3 July 2025 (10.30 am)

Unit weekly activities

Week	Commencing	Topic	Special learning activities	Assessment due date
1	3 March 2025	Introduction to the unit	Defining the design challenge/Problem statement. Introduction to Fusion 360 software.	
2#	10 March 2025	Design for Sustainability	Ideation phase: Generating design ideas, brainstorming and developing the Mind Map.	
3	17 March 2025	Design Analysis	Hand sketching the design concepts. Evaluating alternative designs 3D modelling (using basic tools in Fusion 360)	
4	24 March 2025	Introduction to 3D modelling	3D modelling using pattern tools	
5	31 March 2025	Exploring core tools for 3D modelling	Information sketch 3D modelling using revolve and loft tools	**Assessment 1: Design Portfolio Part 1** due Friday 4 April

6	7 April 2025	Exploring advanced tools, and parametric modelling	Creating shell, and coil features Surface modification tools Equation-derived curve	
7*~	14 April 2025	Conventions for drawing documentation in accordance to AS1100 standards	Create a detailed technical drawing that includes precise dimensioning, section views, and linear and geometric tolerancing. Create an assembly drawing with a Bill of materials (BoM).	
8	28 April 2025	Comprehending component Functionality and product visualisation	Create bodies and components Create joints Exploded view Create realistic images	
9	5 May 2025	Critical design review	Evaluating the design decisions.	
10	12 May 2025	CAD Skills test	N/A	**Assessment 2: CAD Skills test** (Hurdle task) Held in studio session
11	19 May 2025	Summary	Open studio session for assignment completion	
12	26 May 2025			Study period **Assessment 1: Part 2** due Monday 26 May
13-14^	2 June 2025			End-of-unit assessment and exam period

Labour Day public holiday: **Monday 10 March 2025** - University closed
* Easter/Intra-trimester break: **Friday 18 April - Sunday 27 April 2025** (between weeks 7 and 8)
~ ANZAC Day public holiday: **Friday 25 April 2025** - University closed
^ King's Birthday public holiday: **Monday 9 June 2025** - University closed